STITCHER

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A Final Year Project Report Submitted to Abdul Wali Khan University, Mardan in Partial Fulfillment of the Requirement of the Degree BACHELOR OF COMPUTER SCIENCE (BCS)

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FINAL YEAR PROJECT REPORT APPROVAL

This Final Year Project Report Titled

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DEDICATION

We dedicate this thesis to all of the people who have continuously encouraged and supported us academically. To our family, who have supported us in our pursuit of knowledge with their unwavering love, encouragement, and sacrifices? Your confidence on us has served as a constant source of inspiration, and we appreciate your persistent backing. To our mentors and advisors, whose advice, know-how, and priceless insights have impacted our academic and scientific development? Your guidance has encouraged us to push the boundaries of our thinking and pursue excellence. To our mates, thank you for being supportive and understanding in both happy and stressful times. Your companionship has made our life more enjoyable and balanced by serving as a constant reminder of the value of relationships and shared experiences. To the attendees of our study, who generously contributed their time, knowledge, and experiences. Your willingness to participate has enriched this research, and we are grateful for your invaluable contributions. To the academic community, for fostering an environment of intellectual curiosity and collaborative learning. Your collective dedication to the pursuit of knowledge has inspired us to push boundaries and contribute to the advancement of our field.

DECLERATION

We hereby sincerely certify that the "Stitcher" project we submitted for the BS CS at AWKUM is an original piece of writing. We certify that the work we've done here is entirely original it wasn't copied from another source or submitted for academic or professional use elsewhere. We agree that any outside materials used in the creation of this project, including books, research articles, and other materials, have been properly cited and referenced. The appropriate sections of the project report have duly acknowledged any contributions or help received from people or organizations. We further affirm that this project was created with the utmost honesty, integrity, and faithfulness to academic and research integrity norms. Legal duties, moral obligations, or guidelines related to data collection, privacy, and confidentiality have been adhered to throughout the project.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude and appreciation to all those who have contributed to the successful completion of our project titled "Stitcher". Their support, guidance, and encouragement have been invaluable throughout this journey. We extend our deepest thanks to our project supervisor Dr. Faisal Javid for their unwavering support, expertise, and invaluable guidance. Their insightful feedback, constructive criticism, and continuous encouragement have been instrumental in shaping this project. Finally, we would like to express our deepest appreciation to our family for their unconditional love, unwavering encouragement, and understanding throughout our academic journey. Their constant support and belief in my abilities have been our greatest motivation.

ABSTRACT

A flutter-based application called The Stitcher is intended to simplify and automate tailor shop management. It offers a complete platform for tailors and customers to communicate and work together on managing orders and customizing garments. The system provides a user-friendly interface that makes it easier to track orders, place bespoke clothing orders, and manage tailoring jobs. There are many different features and functionalities included in the Stitcher application. Customers can register for personalized accounts, look through a selection of clothing designs, enter measurements, and choose their preferred fabrics. Through integrated messaging systems, they may communicate with tailors in real-time, ensuring clear communication and minimizing misunderstandings. Additionally, customers may follow the status of their orders and get notifications when the status changes, ensuring an efficient and open workflow. Benefits to tailors the Stitcher by gaining access to a centralized dashboard that allows them to manage customer orders efficiently. They can view and accept orders, track measurements, record fabric details, and schedule appointments. The system provides tools for a comprehensive order history, enabling tailors to streamline their operations and enhance customer satisfaction.

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LIST OF ACRONYMS

DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
RES	Resource
RAM	Random Access Memory
SRS	Software Requirement Specification
SDLC	Software Development Life Cycle
UIs	User Interfaces
UML	Unified Modeling Language

CHAPTER 1

INTRODUCTION

1.1 General Background

As innovation develops, we are additionally moving towards specialized kinds of techniques.Nowadays, as an online website and application having a lot of revolution in our life. And we introduced an online website and application that name is “Stitcher”. This application design to provide the multiple technicians Selection of Tailor, measurements, online order booking, unstitched cloth etc. A tailor person whose occupation is making cloths, pants, suits, trousers and jackets etc., to fit individual customers. This application contains different features like Selection of Tailor, measurements, etc. This application is used to expand company’s services in the city digitally so that it has an online presence as well as physical presence. The website would allow end users to choose a Tailor by sitting at home within few clicks on this platform. User can simply scroll through this website and choose the one which suits him, he can use Map navigation to locateall the available tailor near him. Idea of this Website was that you tailor and customer’s use this platform accordingly own needs [1].

1.2 Applications

There are many applications of our project some of the applications of a Stitcher Application are :

1.2.1 Order Management

- Efficient creation and tracking of customer orders.
- Real-time updates on order status and progress.
- Measurement storage for quick reference.
- Automated notifications to customers regarding order updates.

1.2.2 Appointment Scheduler

- Online booking for customers with preferred date and time selection.
- Availability management to avoid double bookings.
- Reminders and notifications for scheduled appointments.

- Calendar integration for easy scheduling and management.

1.2.3 Fabric Inventory and Tracking

- Management of fabric and accessory inventory.
- Tracking availability and quantities of different fabrics.
- Streamlining the procurement process for new fabrics.
- Maintaining fabric cost and supplier information.

1.2.4 Customer Database

- Database for storing customer profiles and contact details.
- Recording customer measurements and preferences.
- Tracking order history and customer preferences for personalized service.

1.2.5 Financial Management

- Invoice generation for orders, including services and material costs.
- Tracking revenue and expenses for financial analysis.
- Generating reports on sales, profits, and business performance.

1.3 Problem Statement

The traditional craft of stitching has always taken a lot of time and effort, and it frequently requires a high level of ability and knowledge to produce intricate and beautiful patterns. However, since digital technology has developed and there are now more online stitching programmers available, more people are now using these platforms to make stitching more available and practical. Online stitching apps provide many advantages, but there are still a number of issues that need to be resolved. Among them include the challenge of locating high-quality stitching patterns, the constrained customizing choices offered by some platforms, and the unfriendly user interfaces that can make it challenging for beginners to get started. The whole experience may also be hampered by platform compatibility problems, software updates, and technological difficulties. User experience. As such, there is a need for a stitching application that addresses these challenges and provide users with a seamless, enjoyable, and engaging experience.

1.4 Proposed Solution

For Men/Women individuals or groups, the Stitcher application will offer a wide range of services required for the sewing. A tailor's services include designing and tailoring clothing such as suits, shirts, trousers and trousers. The goal of the online stitching application is to digitize the order-taking and tailoring processes. They would be able to regularize their daily wagers and save time and work by doing this. This application has a variety of features, including measurements, online order booking, and tailor selection. It must have both a physical and online presence so that clients may make their own decisions. This website is intended to digitally expand the company's services in the city so that it has both a physical presence and an online presence. Thissite would allow end users to choose a Tailor by sitting at home within few clicks on it. Usercan simply scroll through them and choose the one which suits him, he can use Map navigationto locate all the available facilities [3].

1.5 Aims and Objectives

The aims and objectives of Stitcher Application are:

Improve Operational Efficiency: The primary aim of a tailor management system is to enhance the operational efficiency of a tailor's business by automating various tasks and streamlining processes. The system aims to optimize order management, appointment scheduling, inventory tracking, and overall workflow to save time and reduce manual errors.

Enhance Customer Experience: The tailor management system aims to provide an exceptional customer experience by offering convenient online booking, real-time order tracking, and timely notifications to keep customers informed about the status of their orders. The system aims to improve communication, responsiveness, and overall satisfaction of customers.

Increase Productivity and Time Management: The system aims to increase productivity by automating routine tasks, such as appointment scheduling, measurement storage, and order tracking. By providing a centralized platform to manage various aspects of the tailoring business, it enables tailors to allocate their time effectively, focus on core activities, and serve more customers efficiently.

Optimize Inventory Management: The tailor management system aims to improve inventory management by accurately tracking fabric and accessory availability. It helps tailors maintain optimal stock levels, avoid shortages or excesses, and streamline the procurement process. By optimizing inventory, the system aims to reduce costs, improve resource utilization, and minimize delays in order fulfillment.

Facilitate Data Analysis and Decision Making: The system aims to provide comprehensive reporting and analytics capabilities to help tailors gain insights into their business performance. It aims to generate reports on order trends, customer preferences, sales analysis, and financial data. This information assists tailors in making informed decisions, identifying growth opportunities, and implementing effective business strategies [4].

1.6 Project Scope

The project scope of a Stitcher application would typically include the following:

1.6.1 Customer Management

The RES should be able to maintain a customer database that includes customer information such as contact details, measurements, order history, and preferences.

1.6.2 Order Management

The system should enable the tailor to manage orders, including creating new orders, updating order status, and tracking order progress.

1.6.3 Measurement Management

The RES should allow the tailor to enter and store customer measurements and create measurement profiles for each customer. This would enable the tailor to quickly access and use customer measurements for new orders.

1.6.4 Inventory Management

The RES should enable the tailor to manage inventory levels, including tracking the availability of fabrics, trims, and other materials. This would ensure that the tailor can quickly and easily access the required materials for each order.

1.6.5 Reporting

The RES should provide comprehensive reporting capabilities, including reports on order status, inventory levels, customer data, and financial data. These reports can be used to analyze business performance and make informed decisions.

1.6.6 User Management

The RES should have role-based user management capabilities that enable the tailor to control access to system functionality based on user roles.

1.7 Summary

The Stitcher application is a web-based platform created to automate and streamline the tailoring services process. The options it offers include choosing a tailor, getting measurements, and placing an order online. The application intends to promote productivity and time management for tailors, optimize inventory management, improve operational efficiency by automating operations, improve customer experience through easier booking and real-time updates, and facilitate data analysis for informed decision-making. Customer management, order management, measurement management, inventory management, reporting, and user administration are all included in the project's scope. In general, the Stitcher app aims to give customers a seamless and interesting experience in the world of tailoring.

CHAPTER 2

OVERVIEW OF THE EXISTING SYSTEM

Before a new system or technology is implemented, the old system alludes to the current procedure or method that is in use. The overview of the current system in the context of the Stitcher application would entail knowing how customizing services are now provided without the usage of the online platform. Customers looking for tailoring services would often visit real tailor shops or deal directly with individual tailors under the old system. Measurements, fabric selection, ordering, and discussing design preferences are all possible steps in the process. When employing paper-based methods, tailors would manually enter customer information, measurements, and order details. Face-to-face contacts or phone calls would be the primary means of communication between the tailor and the client.

2.1 Manual System

The traditional approach to managing a tailor shop involves using pen and paper to track orders, measurements, inventory, and customer data. This method is time-consuming and prone to errors, and may limit the ability to track and analyze data effectively [5].

2.1.1 Advantages

- Low cost
- Flexibility
- Familiarity
- Personal touch
- Security

2.1.2Disadvantages

- Time-consuming
- Prone to errors
- Limited scalability
- Limited tracking and reporting capabilities
- Inefficient and may slow down business operations

2.2 Spreadsheets

Many tailors use spreadsheets to manage their business operations, including customer data, order details, and inventory. While spreadsheets can provide some level of organization and tracking, they are often not integrated, may require manual data entry, and may lack the ability to provide real-time reporting [6].

2.2.1 Advantages

- Low cost
- Easy to use
- Customizable
- Real-time tracking and reporting
- Multiple users can access and edit the same file simultaneously

2.2.2 Disadvantages

- Limited functionality compared to tailor-specific software
- Prone to errors, such as accidental data deletion or incorrect data entry
- Lack of automation
- Limited scalability
- No centralized database and data may be spread across multiple files, making it difficult to manage and analyze data effectively.

2.3 Off-the-Shelf Software

There are several off-the-shelf software solutions available for tailor management, such as Tailor Pad, Tailoring Solution, and Master Tailor. These software solutions provide features such as order management, inventory management, and billing and payment management. However, these systems may lack customization options and may not be tailored to the specific needs of each tailor shop [7].

2.3.1 Advantages

- Ready-to-use software with a range of features
- Easy to set up and use
- Time-saving
- Integration with other software systems
- Regular updates and support

2.3.2 Disadvantages

- Limited customization options
- May not meet specific needs of a tailor shop
- Lack of flexibility
- High initial investment cost
- Incompatibility with existing software and hardware infrastructure.

2.4 Cloud-based Software

Cloud-based software, such as Trello, Asana, and Evernote, provides the ability to manage tasks, appointments, and notes across multiple devices and locations. These tools can be useful for tailors who need to manage their operations remotely or collaborate with team members in different locations [8].

2.4.1 Advantages

- Accessible from anywhere, anytime with internet connection
- Cost-effective and scalable
- Centralized data storage and management
- Easy collaboration and sharing of data across multiple devices and locations
- Automatic updates and backups.

2.4.2 Disadvantages

- Dependence on internet connection
- Security concerns, such as data breaches and hacking
- Limited control over data privacy and security measures
- Lack of customization options
- Subscription-based pricing model can be expensive over time.

2.5 Mobile Applications

Some tailors use mobile applications such as Tailor Master, Tailor App, and TailorSoft to manage their business operations on the go. These applications provide features such as order management, measurement management, and customer data management, all accessible from a mobile device. However, these applications may not provide the same level of functionality as desktop-based software and may be limited by screen size and device compatibility [9].

2.5.1 Advantages

- Convenient and accessible on-the-go
- Mobile-friendly interfaces optimized for smaller screens
- Real-time updates and notifications
- Easy access to customer data and order management from anywhere
- Integration with mobile device features, such as cameras and GPS.

2.5.2 Disadvantages

- Dependence on mobile devices and operating systems
- Limited functionality compared to desktop software
- Security concerns, such as data breaches and hacking
- Compatibility issues across different mobile devices and platforms
- Additional development costs for tailored app features.

2.6 Summary

The current management system for tailor shops consists of hand-written methods, spreadsheets, commercial software, cloud-based software, and mobile apps. The manual system is time-consuming and error-prone, but it is inexpensive. Spreadsheets provide some organization but lack real-time reporting and integration features. Although ready-to-use capabilities are offered by off-the-shelf software, customization choices may be limited. Although it requires an internet connection, cloud-based software enables remote access and collaboration. Although they may have limited functionality and compatibility, mobile applications are convenient. Overall, there are limitations to the current system's effectiveness, scalability, tracking capacities, and data processing.

CHAPTER 3

TOOLS AND METHODOLOGY

The terms "tools" and "methodology" relate to the particular applications, methods, and strategies used to support and direct the creation or deployment of a project or system. The software and methods used to design, create, and administer the Stitcher application would fall under the category of tools and methodology.

3.1 Requirement Analysis

Toward the beginning of the necessary investigation process, the normal deliverable is the software requirement specification (SRS), which depicts the ways of behaving and highlights the system. It additionally contains a rundown of the fundamental of the framework along with the graphs. The fundamental list is the rundown of capabilities that a system should have. Fundamental are of two classes, which are utilitarian and nonfunctional. Useful prerequisites are the necessity that the partners need from the system, how is everything turning out to be worked, and what the system ought to have. Non-practical necessities are the prerequisites that determine the models that can be utilized to decide how the framework works. The graphs, which are otherwise called the system model are the deliberation of the system. Each model is introducing a specific perspective on the system. The graphs additionally show the connection between the system and the outer substances [10].

3.1.1 Functional Requirement

The prerequisite investigation is a computer programming approach that comprises a progression of exercises that lay out the requests or conditions that should be fulfilled for a new or refreshed item while considering the potential for contending necessities from various clients. Practical prerequisites are those that are utilized to show the framework's inward-working nature, as well as the framework's portrayal and clarification of every subsystem. It includes the errand that the framework ought to achieve, the cycles in question, the information that the framework ought to contain, and the UIs [11]. The useful prerequisites are shown in Table 3.1.

Table 3. 1 Functional Requirement

Req. No.	Description	Type
R-1	A user should be able to register by email.	Functional
R-2	A user should be able to Monitor.	Functional
R-3	The system allows the admin to delete the user.	Functional
R-4	The system should display the summary of the current user.	Functional
R-5	Admin should be able to log in to the system using his/her email and password.	Functional
R-6	The system should allow new user accounts to be created by the admin	Functional

3.1.2 Non-Functional Requirement

It portrays framework components that are worried about how the framework satisfies practical necessities. They are as per the following:

1) Security

Just approved corporate laborers might gain admittance to the company's gotten page on the frameworks, and just clients with appropriate passwords and usernames can sign in to see the client’s page.

2) Performance and Response Time

The framework ought to have an elite exhibition rate while executing client input and ought to have the option to offer criticism or a reaction in a short measure of time, frequently 50 seconds for very troublesome exercises and 20 to 25 seconds for less refined positions.

3) Error Handling

Mistakes ought to be kept away from however much as could be expected, and a reasonable blunder message ought to be provided to assist the client through the recuperation with handling. The significance of approving client input couldn't possibly be more significant. Moreover, the time it takes to recuperate from a slip-up ought to be somewhere in the range of 15 and 20 seconds.

4) Availability

This framework should be open consistently, 24 hours every day, seven days per week. In case of a horrendous framework disappointment, the framework ought to be back ready within 1 to 2 work days, guaranteeing that the business cycle isn't upset.

5) Ease of Use

Given the shopper's degree of understanding, a fundamental yet top-notch UI ought to be made to simplify it to fathom and need insignificant preparation [12].

Table 3. 2 Non-Functional Requirements

Req. No.	**Description**	**Type**
R-1	The system should be trusted and relied on by the user.	Reliability
R-2	The system should be easy for the customers to use it.	Usability
R-3	The System should run on any hardware with any kind of browser. It should not conflict with other processes within this environment.	Portability and Compatibility
R-4	The system Should keep running when it launches unless there is an intentional shutdown of the system.	Performance

3.2 Software Requirement

The product prerequisites are a depiction of elements and functionalities of the objective framework. Prerequisites convey the assumptions for clients from the product item. The prerequisites can be self-evident or covered up, known or obscure, expected or unforeseen according to the client's perspective. Programming necessity is a practical or non-utilitarian should be executed in the framework. Practical means offering specific support to the client. Programming necessity can likewise a nonfunctional, it tends to be an execution prerequisite. For instance, a non-practical necessity is where each page of the framework ought to be noticeable to the clients in 5 seconds or less [13].

3.3 Hardware Requirement

The hardware requirements for this project will depend on the specific features and functionality of the system, but in general, the following hardware components may be required:

Computer or Server: A computer or server will be required to run the software and store the data associated with the construction project.

Database: A database will be required to store and organize the data associated with the construction project, such as project schedules, budgets, and resource allocations.

Mobile Devices: Some systems may allow for access to the project from mobile devices such as smartphones and tablets, so a mobile device with internet access and compatible software may be required.

Backup and Recovery Systems: To ensure data integrity and availability, backup and recovery systems may be required to store and protect the data associated with the construction project.

3.4 Software Tools

The run time climate is the climate wherein a program or application is executed. The equipment and programming foundation upholds the running of a specific codebase progressively. Coming up next are the product necessities of our undertaking.

- Android Studio
- Flutter
- Google Chrome or any other browser

3.5 Hardware Tools

The hardware required to run a specific software will depend on the requirements of that software. However, in general, the following hardware components may be required:

Computer: A computer is required to run the software and store any associated data. The minimum specifications for the computer or server will depend on the requirements of the software, such as the amount of memory and processing power required.

Operating System: An operating system is required to run the software, such as Windows, MacOS, or Linux. The software must be compatible with the operating system installed on the computer or server.

Memory: The software may require 1GB amount of memory (RAM) to run effectively.

Storage: The software may require a certain amount of storage space on the computer or server to be installed and run.

Display: If the software has a graphical user interface, a display is required to view the interface and interact with the software.

Input Devices: The software may require input devices such as a keyboard or mouse to interact with the software.

Networking: Some software may require network connectivity to access and transfer data or communicate with other devices.

3.6 Summary

The above chapter provides an overview of the requirements analysis process for the Stitcher application, including functional and non-functional requirements. It discusses the software and hardware requirements for the project, as well as the tools and methodology to be used. The functional requirements include user registration, monitoring, and user deletion, display of user summaries, admin login, and user account creation. The non-functional requirements encompass aspects like security, performance, error handling, availability, and ease of use. The software tools mentioned include Android Studio, Flutter, and web browsers, while the hardware tools depend on the specific software requirements and may include computers, operating systems, memory, storage, displays, input devices, and networking capabilities.

CHAPTER 4

SYSTEM MODELING AND DESIGN

Once the requirements document for the software to be developed is available, the software design phase begins. While the requirement specification activity deals entirely with the problem domain, design is the first phase of transforming the problem into a solution. In the design phase, the customer and business requirements and technical considerations all come together to formulate a product or a system. Software design is a process to transform user requirements into some suitable form, which helps the programmer in software coding and implementation. Software design is the first step in SDLC, which moves the concentration from the problem domain to the solution domain. The software design of Stitcher Application is actually to specify to fulfill the requirements mentioned in SRS. The purpose of the System Design process is to provide sufficient detailed data and information about the system and its system elements to enable the implementation consistent with architectural entities as defined in models and views of the system architecture. We used different types of techniques to bring about the design of a system like a system diagram, Data Flow Diagram, and entity-relationship diagram [14].

4.1 Context Diagram

The Context diagram is a visual model of a Stitcher Application, its components, and their interactions, and it can capture all the essential information of a system’s design. The context diagram is used to establish the context and boundaries of the system to be modeled: which things are inside and outside of the system being modeled, and what is the relationship of the system with these external entities. Also, a context diagram can be used to reduce risks in a project greatly. Figure 4.1 represents the visual model of Stitcher Application the end-user can perform all the basic operations of Stitcher Application.

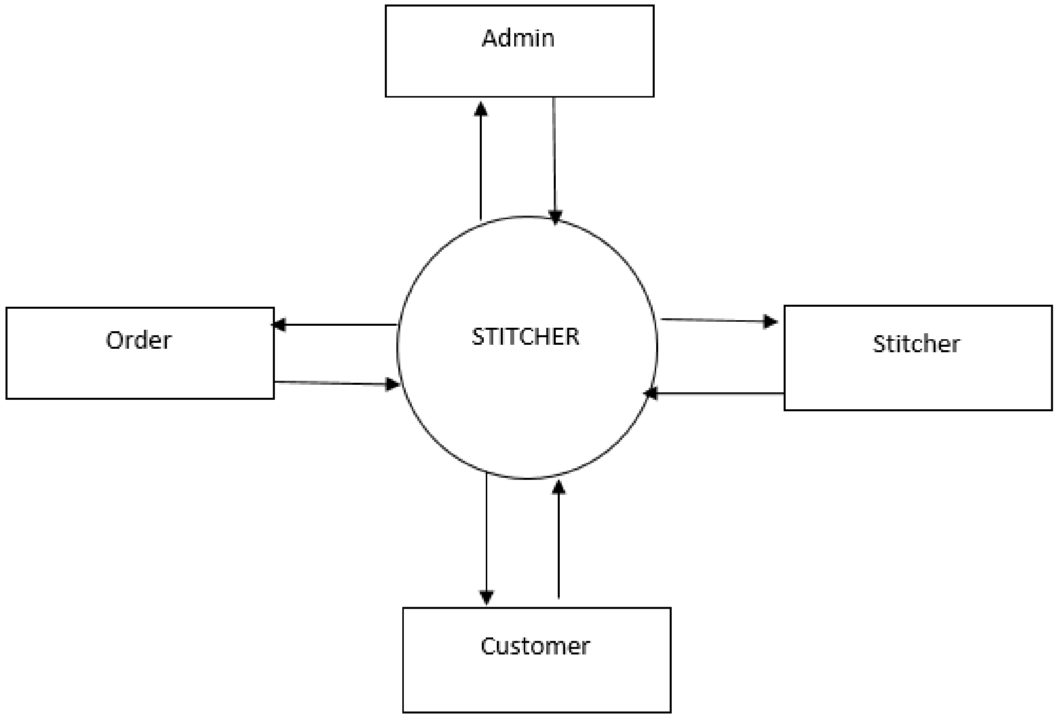

Figure 4. 1 Context Diagram

4.2 Use Case Diagram

Use-case diagrams describe the high-level functions and scope of a system. These diagrams also identify the interactions between the system and its actors. The use cases and actors in use-case diagrams describe what the system does and how the actors use it, but not how the system operates internally. Figure 4.2 shows the use case diagram.

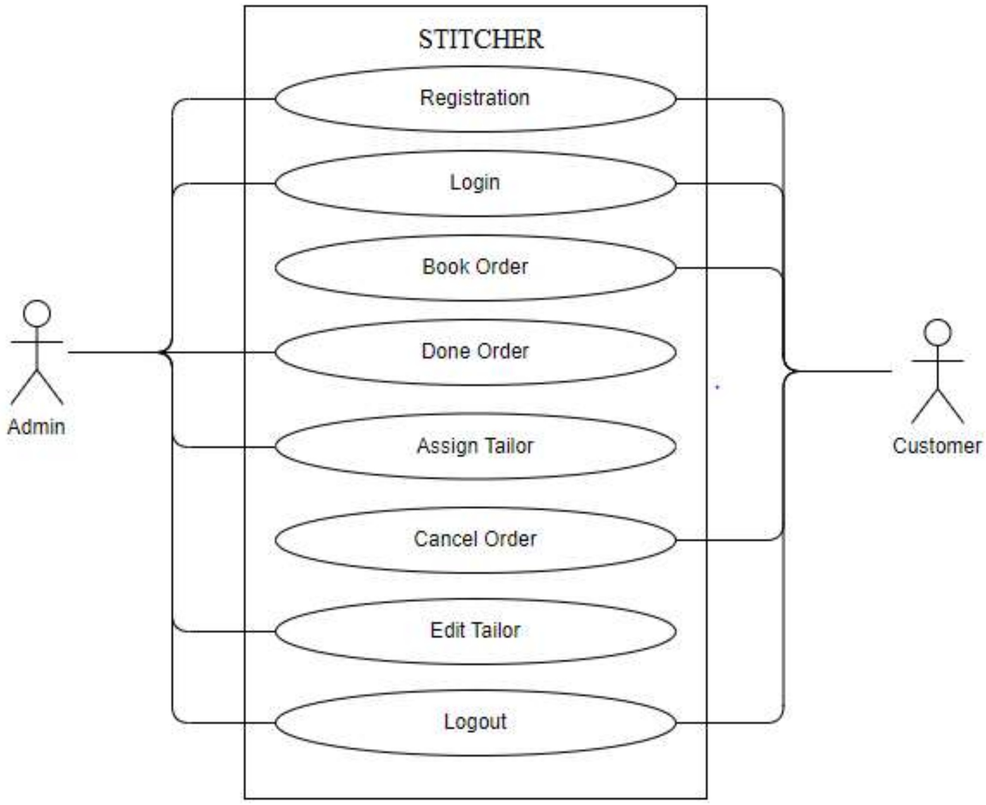

Figure 4. 2 Use Case Diagram

4.3 Activity / Sequence Diagram

In UML Sequence Diagrams are interaction diagrams that detail how operations are carried out. They capture the interaction between objects in the context of a collaboration. Sequence Diagrams are time focus and they show the order of the interaction visually by using the vertical axis of the diagram to represent time what messages are sent and when. Figure 4.3 shows the sequence diagram of and Stitcher Application.

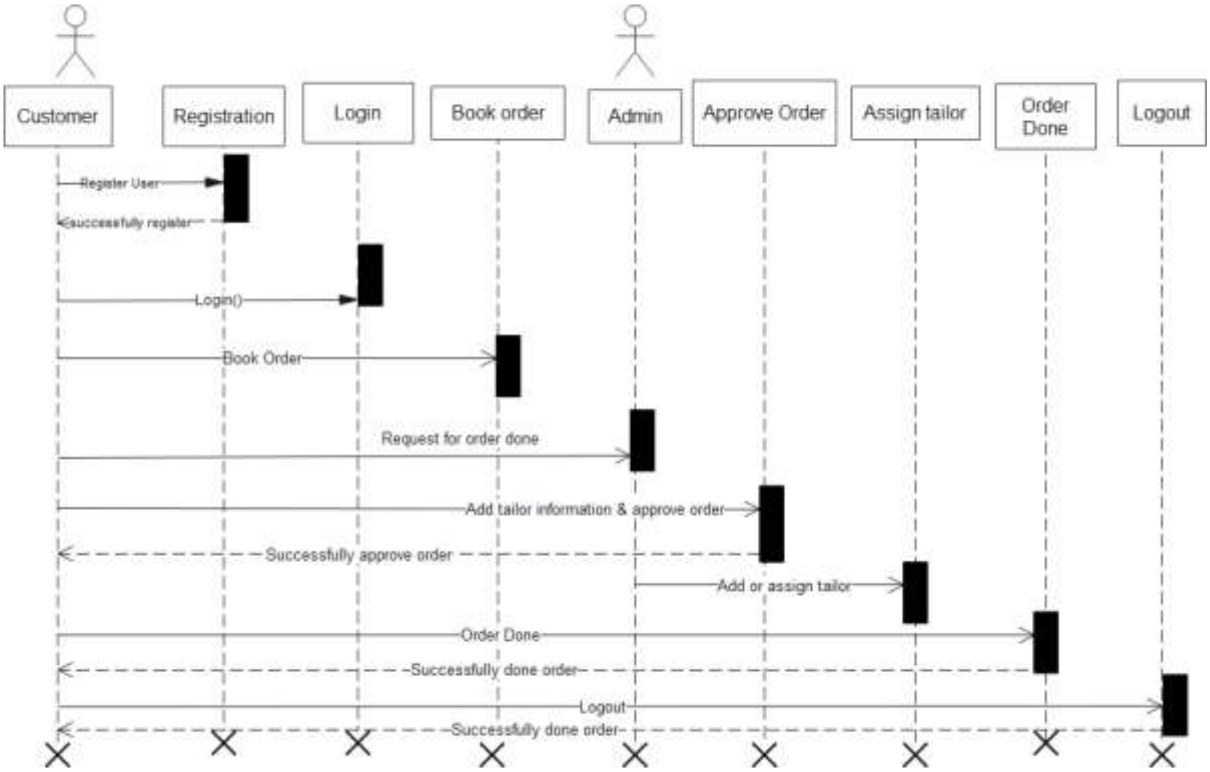

Figure 4. 3 Sequence Diagram

4.4 System Architecture

System architecture refers to the overall structure and design of a software system or application. It defines the arrangement of components, modules, and subsystems, as well as the interactions and relationships between them. The system architecture provides a blueprint that outlines how different parts of the system work together to achieve the desired functionality and performance. In a Stitcher Application, the system architecture defines how various modules and functionalities are organized and integrated to support the overall objectives of the system.

In the following figure 4.4 when the user launch the application the user must have to select tailor before giving measurements. The user will give measuememt details and book te order. The user can also track his order.

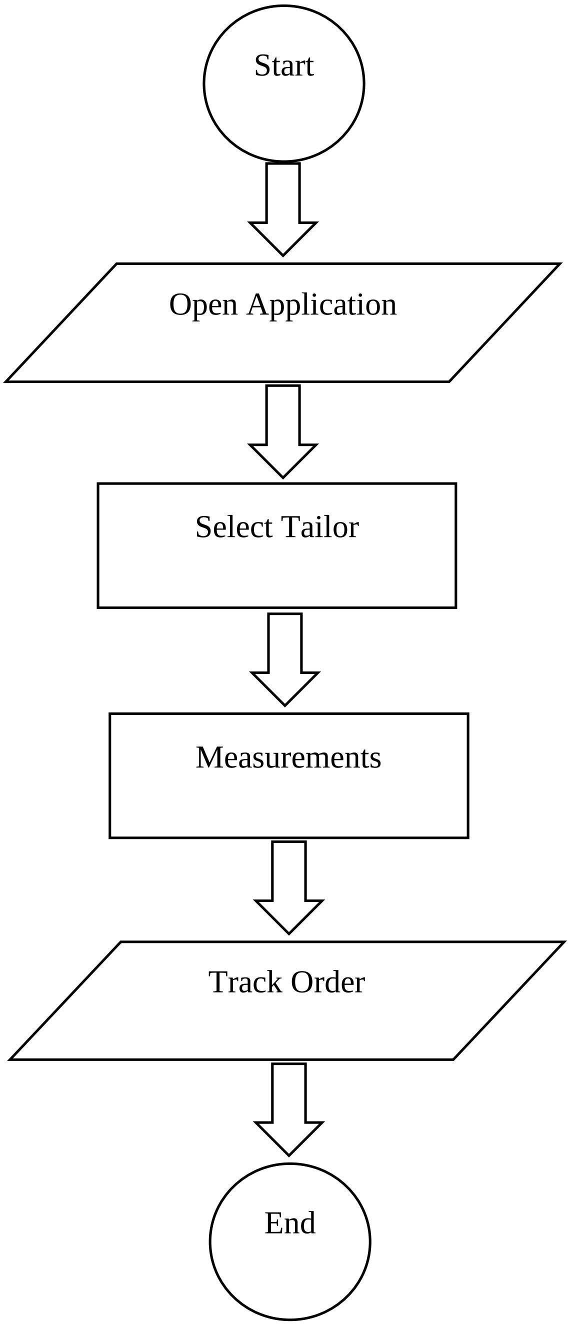

Figure 4. 4 System Architecture

4.5 Data Flow Diagram

A data flow is a representation of the data and relationships within a system. In a Stitcher, the data model typically includes entities such as customers, products, orders, and payments, as well as their relationships to each other. It also includes information about the attributes and constraints of each entity, such as customer name, product price, and order status. The data model is used to define the structure of the data and ensure consistency and integrity of the data within the system.

In figure 4.5 the data will flow through customers, stitchers, and booking orders. The customer will select tailor and book the order.

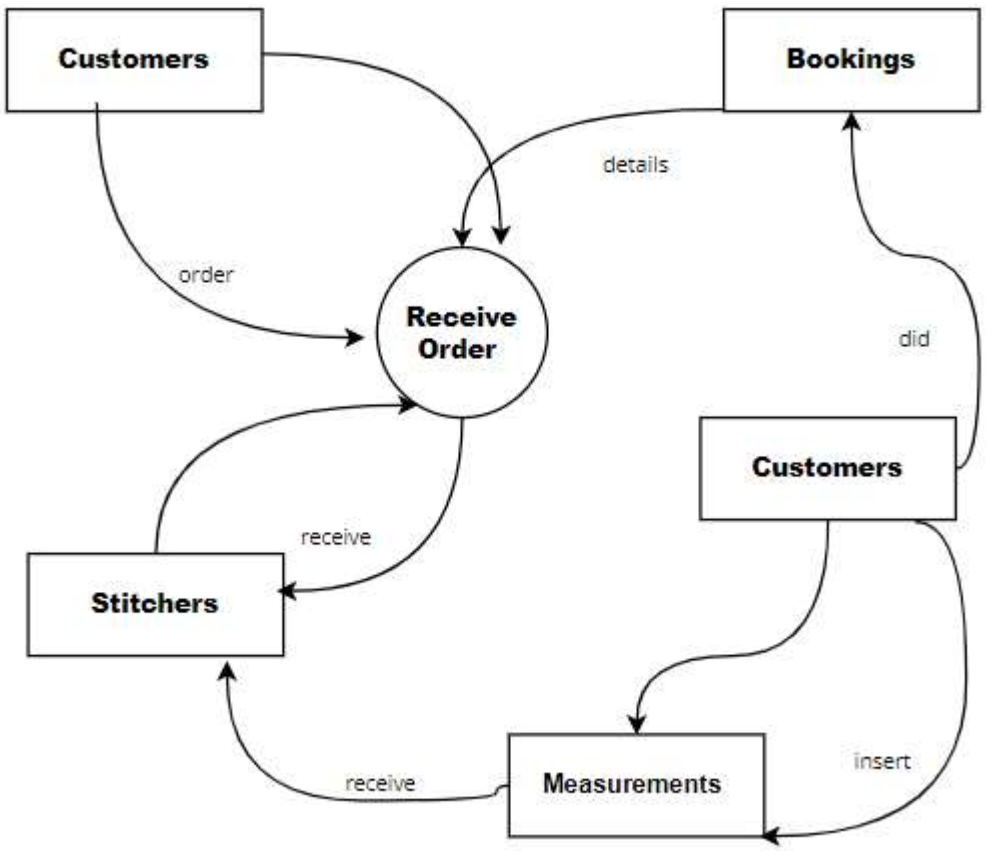

Figure 4. 5 Data Flow Diagram

2.6 Entity Relationship Diagram

An Entity-Relationship Diagram (ERD) is a visual representation that depicts the relationships between different entities in a database. It is a powerful tool used in database design and development to model the structure and relationships of data entities. In an ERD, entities are represented as rectangles, and the relationships between entities are depicted using lines connecting them. Each entity represents a specific object or concept in the database, such as a customer, order, or product. Relationships between entities illustrate how these objects are related to each other and provide insights into the data dependencies and associations within the database. The figure 4.6 shows the ERD of our proposed system.

- **Admin entity:** Represents an administrator of the tailor management system. It has attributes like admin_id (unique identifier for the admin), username, password, name, email.
- **Customer entity:** Represents a customer who wants to place orders for tailoring services. It has attributes like customer_id (unique identifier for the customer), name, email, city, country, gender and contact.
- **Tailor entity:** Represents a tailor who provides tailoring services. It has attributes like tailor_id (unique identifier for the tailor), name, email, contact, country, lat, lang, and image.
- **Measurement entity:** Represents the measurements taken for a customer's order. It has attributes like measurement_id (unique identifier for the measurement), customer_id (to identify which customer the measurement belongs to), tailor_id (to identify which tailor took the measurement), chest, tira, bazo, gera, and kaff.
- **Order entity:** Represents an order placed by a customer. It has attributes like order_id (unique identifier for the order), customer_id (to identify which customer placed the order), tailor_id (to identify which tailor is handling the order), date, and time.

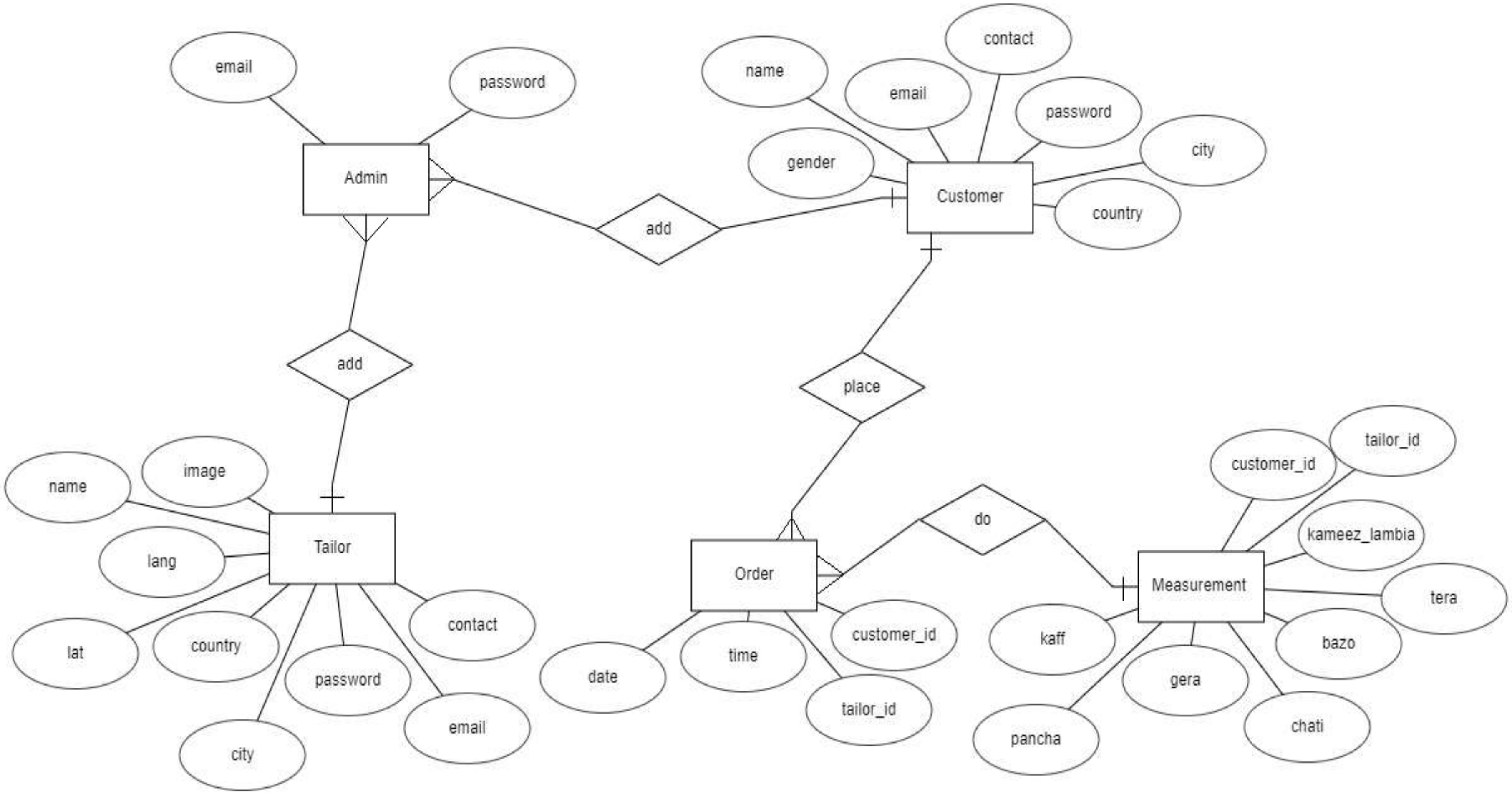

Figure 4. 6 ER Diagram

CHAPTER 5

IMPLEMENTATION AND TESTING

Implementation is the process of turning a software application's concept and specifications into a functional system. It entails writing code, creating the required elements and functionalities, integrating various modules, and setting up the system in accordance with the specifications. Developers construct the application using programming languages, frameworks, and tools in accordance with the requirements and design standards.

5.1 User Interfaces

5.1.1 Admin Login

The Login screen appear to the admin when the admin open admin panel. The admin should require email, password to enter to the admin panel. Admin login screen are in the figure 5.1.

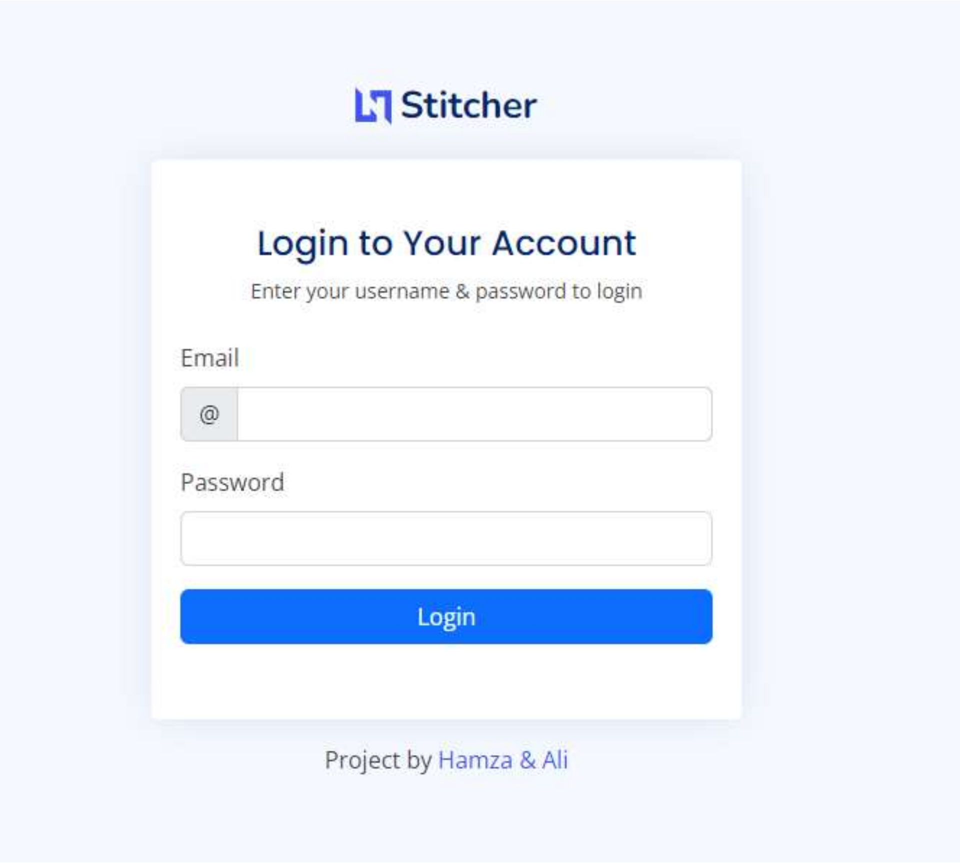

Figure 5. 1 Login Screen

5.1.2 Admin Dashboard

The screen are the admin dashboard screen which show all the record to the admin. The screen shows the information about Customers, tailor, Expenses, Orders to the admin. Admin dashboard screen shows in the figure 5.2.

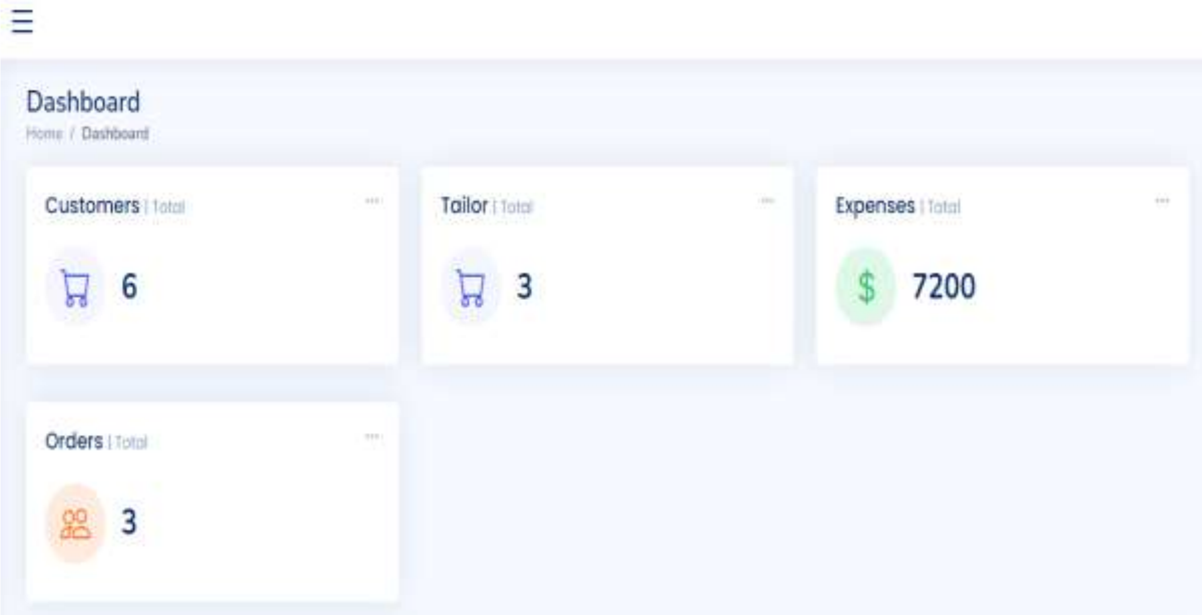

Figure 5. 2 Admin Dashboard

5.1.3 Tailor List

The screen appear to the admin when the admin want to see all the tailor registered to the application. The admin also manage all the tailor record like add, update, delete etc. Figure 5.3 shown the tailor list screen.

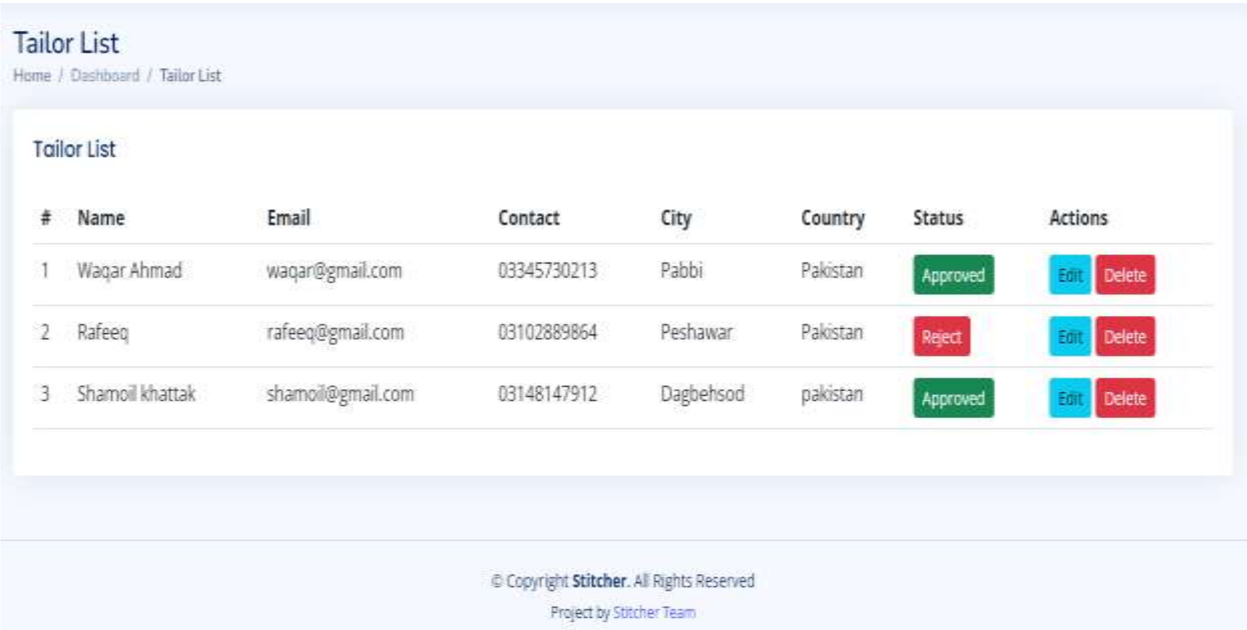

Figure 5. 3 Tailor List

5.1.4 Order List

The screen are the order screen. The screen appear to the tailor when he want to see all orders of customers. The order list screen are the follow. The figure 5.4 shows the order list screen.

Figure 5. 4 Oder List

5.1.5 User / Customer List

The figure 5.5 are the customer list screen. The screen shows the customer list to the admin which are registered to the application. The User list screen are the follow. The customer list screen are shown in the figure 5.5.

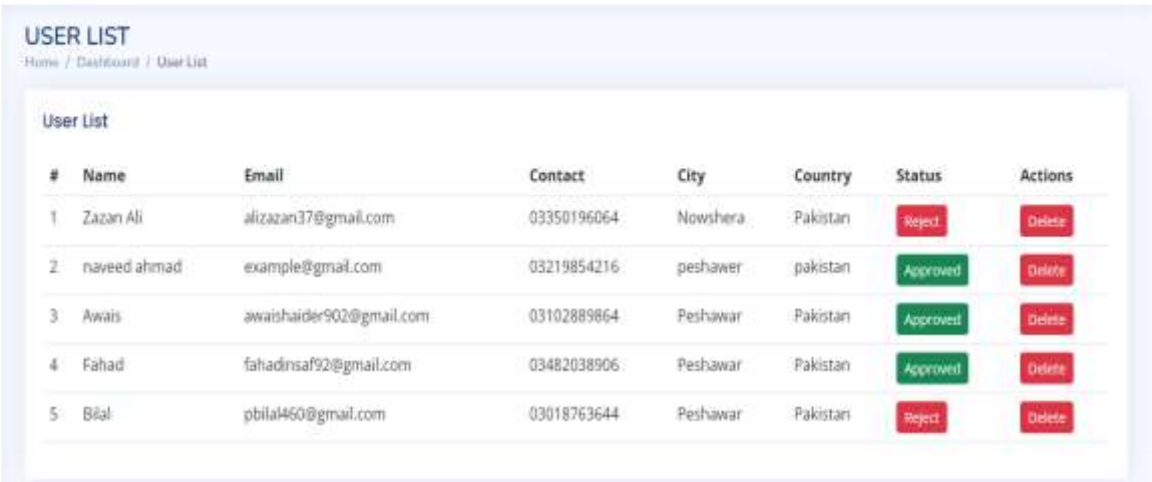

Figure 5. 5 User List

5.1.6 Expenses List

The screen are the expenses list screen. The expense list screen show all the expenses done by tailor in the shop. The Expenses are shown with specific name, amount, details and date to the tailor. The Expense list screen are shown in the figure 5.6.

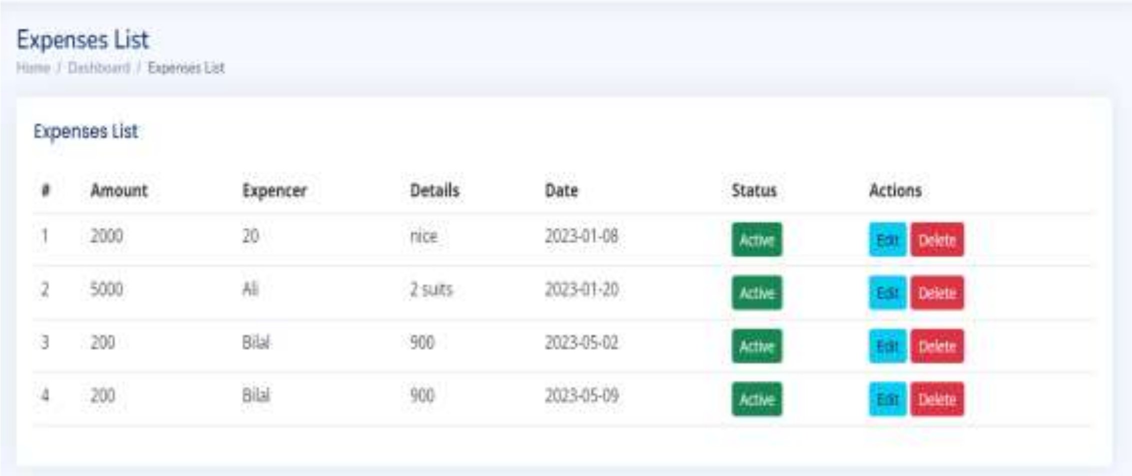

Figure 5. 6 Expenses List

5.1.7 User Login

The screen are the user login screen. The following screen appear to the user when the user open the application. Every user must have to login or register himself before proceed to the application. The Figure 5.7 shows the user list screen.

Figure 5. 7 Customer Login

5.1.8 Customer Dashboard

The customer dashboard of a tailor application serves as a centralized hub where customers can manage their tailoring orders, track progress, and communicate with the tailor. It provides a user-friendly interface that enhances the overall tailoring experience for customers. The figure 5.8 shows the customer list screen.

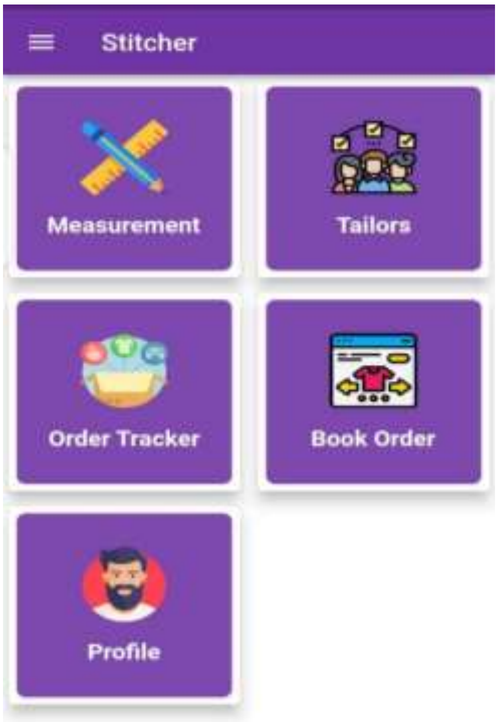

Figure 5. 8 Dashboard Screen

5.1.9 Order Screen

The order screen is a crucial component of a Stitcher application. It serves as the central hub for managing and tracking customer orders, facilitating seamless communication between the tailor and their clients. This screen provides an organized and efficient way to handle various aspects of the tailoring process, ensuring that orders are processed accurately and on time.

When accessing the order screen, the tailor is presented with a clean and intuitive user interface. The screen is typically divided into multiple sections to display different information related to orders. The order screen are shown in the figure 5.9.

Figure 5. 9 Order Screen

5.1.10 Track Order

The track order feature in a tailor application provides customers with real-time updates on the status and progress of their tailored garments. It enables customers to stay informed about the various stages of the tailoring process and ensures transparency and accountability. The following figure 5.10 shows the track order screen.

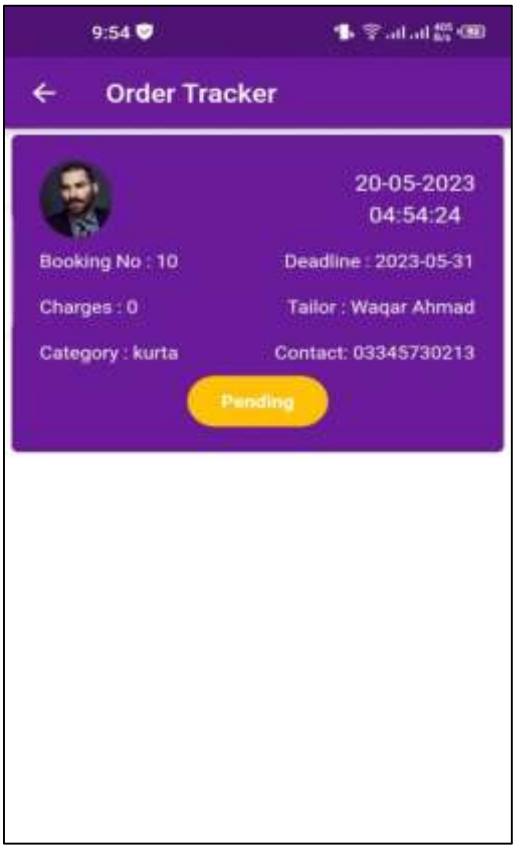

Figure 5. 10 Order Track Screen

5.1.11 Tailor List

The tailor list screen in a Stitcher application is a centralized and organized view that displays a list of tailors registered within the application. It serves as a directory or catalog of tailors, providing essential information about each tailor and facilitating seamless communication and collaboration between the application users and tailors. The tailor list screen are in the figure 5.11

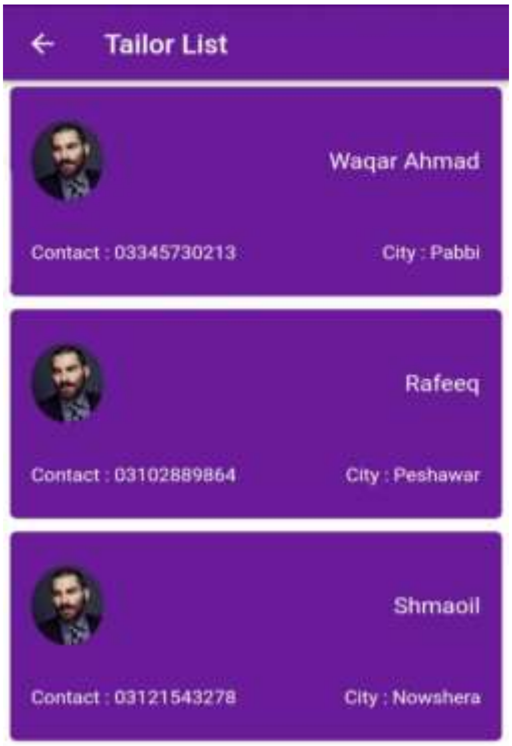

Figure 5. 11 Tailor List

5.2 Software Testing

Software testing is a systematic process of evaluating a software application or system to identify defects, errors, or discrepancies between expected and actual results. It involves verifying and validating that the software meets specified requirements and functions as intended. The primary objectives of software testing are to ensure the quality, reliability, and performance of the software, and to mitigate risks associated with using or deploying it. Through testing, developers and quality assurance professionals can uncover bugs, errors, and vulnerabilities, allowing for timely detection and resolution before the software is released to end-users [15].

5.2.1 Unit Testing

Unit testing is a type of software testing that focuses on testing individual units or components of a software application in isolation. A unit refers to the smallest testable part of the software, such as a function, method, or class.

The main goal of unit testing is to ensure that each unit of code functions correctly and performs as expected. By isolating units and testing them independently, developers can identify and fix defects early in the development process, improving the overall quality and reliability of the software. Unit testing table are shown in table 5.1.

Table 5. 1 Unit Testing

Unit Test Case	Description
createCustomerTest	Tests the creation of a new customer in the Stitcher Application, verifying that the customer is successfully added to the system.
createOrderTest	Tests the creation of a new order in the Stitcher Application, ensuring that the order is successfully added to the system.
calculatePriceTest	Tests the calculation of the price for a given order, verifying that the pricing logic, including fabric, measurements, and design, is accurate.
updateOrderTest	Tests the updating of an existing order, ensuring that the order details can be modified and saved correctly in the tailor management system.
deleteOrderTest	Tests the deletion of an order, verifying that the order is successfully removed from the system without any adverse effects.
trackOrderStatusTest	Tests the retrieval of order status, ensuring that the Stitcher Application accurately retrieves and displays the current status of an order.
generateReportsTest	Tests the generation of reports, such as order statistics, customer insights, or revenue analysis, ensuring that the reports are generated correctly.
createCustomerTest	Tests the creation of a new customer in the Stitcher Application, verifying that the customer is successfully added to the system.

This table provides a structure for documenting unit tests for a Transportation Management System (TMS). The test cases listed above are just placeholders and should be replaced with actual test cases specific to your TMS implementation.

Additionally, you may include additional columns in the table to capture more information such as test data, test environment, and any comments or notes related to the test execution.

5.2.2 Integration Testing

Integration testing is a software testing technique that focuses on testing the interaction and integration between different components, modules, or subsystems of a software application. The purpose of integration testing is to verify that the combined units function correctly as a whole and to identify any issues that may arise due to the integration. Integration testing involves testing the interfaces, data flow, and communication between various components to ensure they work together seamlessly and produce the desired outcomes. It aims to uncover defects or inconsistencies that may occur when different units or modules interact, such as data mismatches, communication failures, or incompatible dependencies. The integration testing table are shown in Table 5.2.

Table 5. 2 Integration Testing

Test Case ID	Description	Description	Steps	Result	Pass/Fail
IT001	Customer Creation Integration	Stitcher Application is running	1. Create a new customer through the user interface.	The customer is successfully created and saved in the system.	Pass
IT002	Order Creation Integration	Stitcher Application is running, customer exists	1. Select a customer from the existing list. 2. Enter order details such as measurements,	The order is successfully created and associated with the selected customer	Pass

			fabric, and design. 3. Save the order.		
IT003	Invoice Generation Integration	Stitcher Application is running, order exists	1. Select an existing order. 2. Generate an invoice for the order	An invoice is generated with accurate order and customer details.	Pass
IT004	Inventory Management Integration	Stitcher Application is running	1. Add a new fabric to the inventory. 2. Update the quantity of an existing fabric. 3. Remove a fabric from the inventory.	The inventory is updated accordingly with the new fabric added, quantity updated, and fabric removed.	Pass
IT005	Reporting Integration	Tailor management system is running	1. Generate a sales report for a specific time period. 2. Generate a customer report with filters.	The sales report and customer report are generated accurately based on the provided criteria.	Pass

The table typically includes a list of test scenarios or test cases that cover various aspects of integration between the TMS and other related systems, such as order management systems, carrier systems, warehouse management systems, or financial systems. Each test scenario represents a specific integration point or interaction between different modules or components of the TMS. For each test scenario, the Integration Testing Table provides details such as the test case ID, a brief description of the scenario, the systems or components involved, the expected input or data, the expected output or result, and any additional notes or requirements for the test execution. This information helps the testing team understand the purpose and scope of each integration test and ensures that all necessary checks are performed.

5.2.3 System Testing

System testing is a type of software testing that is performed to evaluate the behavior and functionality of a complete, integrated system. It is conducted on the entire system as a whole, rather than focusing on individual components or units. The main objective of system testing is to ensure that all the components of the system work together harmoniously and meet the specified requirements. It aims to identify any defects or inconsistencies in the system's behavior, performance, and functionality. During system testing, the software is tested in an environment that closely resembles the production environment to simulate real-world usage conditions. This includes testing interactions between different subsystems, verifying the system's ability to handle inputs and produce expected outputs, and assessing its compliance with functional and non-functional requirements. The System testing are shown in the Table 5.3.

Table 5. 3 System Testing

Test Case ID	Description	Test Steps	Result	Pass/Fail
ST001	Customer Creation	. Access the system with valid credentials. 2. Navigate to the customer creation module.	The customer is successfully created and saved in the system.	Pass

		3. Enter valid customer details and save.		
ST002	Order Creation	Access the system with valid credentials. 2. Navigate to the order creation module. 3. Select an existing customer. 4. Enter valid order details and save.	The order is successfully created and associated with the selected customer.	Pass
ST003	Inventory Management	Access the system with valid credentials. 2. Navigate to the inventory management module. 3. Add a new fabric to the inventory. 4. Update the quantity of an existing fabric. 5. Remove a fabric from the inventory.	The inventory is updated accordingly with the new fabric added, quantity updated, and fabric removed.	Pass

ST004	Reporting Functionality	Access the system with valid credentials. 2. Navigate to the reporting module. 3. Generate a sales report for a specific time period. 4. Generate a customer report with filters.	The sales report and customer report are generated accurately based on the provided criteria.	Pass

The table provides an overview of what each test case aims to verify, and the test steps outline the specific actions to be performed. The expected result specifies the desired outcome, and the pass/fail status indicates whether the test case passed or failed based on the actual outcome observed during testing.

5.3 Summary

In this chapter we shows the implementation and testing of our proposed system Stitcher application. Firstly we insert the screenshots of our project and describe each screenshot. After that we create testing tables for our project in which we test every possible functionality of our proposed system.

CHAPTER 6

SUMMARY AND CONCLUSION

6.1 Summary

A software application developed expressly to meet the requirements of tailoring companies is known as Stitcher. It works as a thorough solution to automate and streamline many facets of running a tailor shop or business, such as customer administration, order monitoring, inventory control, and financial management. Tailors may effectively handle customer data, like as measurements, preferences, and order history, with the help of stitcher application. They can simply track orders, access and update customer profiles, and offer specialized services based on distinct consumer needs thanks to the system. This raises consumer satisfaction and promotes the development of lasting partnerships a crucial component of a tailor's business is order tracking, which a TMS streamlines by offering a centralized platform to track and handle orders from beginning to end. Tasks may be created and delegated, orders can be tracked, and reminders for crucial deadlines or milestones can be sent to tailors. This guarantees effective order processing, reduces errors, and aids in timely order delivery. An additional crucial component of a TMS is inventory control. It enables tailors to monitor their inventory of fabric, trims, and other materials to make sure they have what they need to complete orders. When stock levels are low, the system can provide warnings or messages, allowing for prompt restocking and preventing production delays.

6.2 Conclusion

In conclusion, the System has proven to be a highly effective approach to organizational management, particularly in the realm of manufacturing and production. Developed by Frederick Winslow Taylor in the late 19th and early 20th centuries, this system emphasizes the importance of standardization, efficiency, and the scientific analysis of work processes.

While some critics have argued that the System dehumanizes workers and reduces them to mere cogs in a machine, others have noted that it has helped organizations achieve remarkable gains in productivity and profitability. Moreover, modern variations of the system have incorporated elements of employee empowerment and engagement, while still adhering to the principles of efficiency and standardization.

Overall, the System remains a significant contribution to the field of organizational management, and its principles continue to inform many management practices today. As organizations continue to face the challenges of a rapidly changing business landscape, the enduring insights of Frederick Winslow Taylor may provide valuable guidance for achieving success in the years ahead.

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